High-Level Design – Board

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** High-Level Design (sub-module 02.3 Board, code `BOARD`) · **Version:** 1.0 · **Status:** Draft
> **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026
> **Author:** Jojo George (jojogeorge3344@gmail.com)
> **Based on:** BRD-Board v1.0, FRD-Board v1.0
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Document Control

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version, derived from BRD/FRD-Board v1.0, the verified Kanban Board technical doc, TDD v2.0 and the code

Related documents:

Document	Location
Overall HLD (stack, deployment, cross-cutting design, Auth)	`../../00-Overall/HLD-TaskFlow-Overall.md`
BRD / FRD – Board	`../../../BRD/02-TaskFlow-Monitoring/02.3-Board/BRD-Board.md`, `../../../FRD/02-TaskFlow-Monitoring/02.3-Board/FRD-Board.md`
LLD – Board	`../../../LLD/02-TaskFlow-Monitoring/02.3-Board/LLD-Board.md`
Sibling HLDs – Task Management (owner of the endpoints and rules used here), Employee Workload	`../02.1-Task-Management/HLD-Task-Management.md`, `../02.2-Employee-Workload/HLD-Employee-Workload.md`
Related HLD – Notification (reminder engine)	`../../04-Communication/04.2-Notification/HLD-Notification.md`
TDD	`TDD-TaskFlow.md` (v2.0) §7.5, §10.2, §12

1. Introduction

1.1 Purpose

This HLD describes how the Board (Kanban) is built: its place in the system, its frontend component, the Task Management services it reuses, its flows, security and design decisions. Class- and

endpoint-level detail is in the LLD.

1.2 Scope

In scope: the Board screen (`/board`) – five status columns, cards, drag-and-drop status change, Project / Assignee / Status / Deadline filters, Refresh and opening a card. The server components it uses (task scope, transition table, history) are owned by Task Management and only summarised here.

Change from original BRD (Screen 10)

Original BRD	As built (this design)
Four columns: To Do, In Progress, Blocked, Completed	Five columns, **In QA** added; no Cancelled column
Card shows status	Column gives the status; card adds the project
No filters	Project, Assignee, Status and Deadline filters (client-side)

1.3 Audience

Developer, reviewer and tester of TaskFlow (Jojo George); anyone maintaining the Monitoring module.

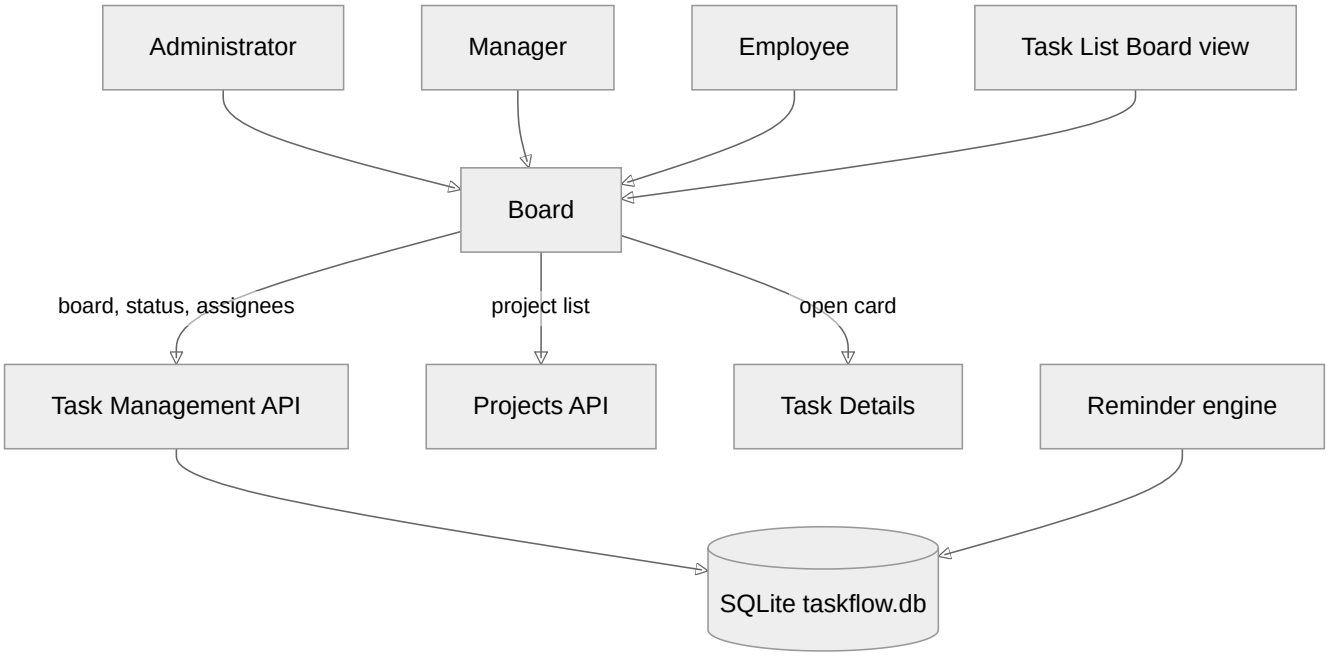
1.4 Definitions & Abbreviations

Term	Meaning
SPA	Single Page Application (Angular frontend)
Card	One task on the board
Optimistic move	The card moves on screen at once and moves back if the server refuses
Overdue card	Deadline before now and task neither Completed nor Cancelled (instant rule)
Task scope	Own, Team or Organization – the same scope as the Task List
CDK	`@angular/cdk`, the Angular Component Dev Kit (drag-and-drop)

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2. System Context

2.1 Context Diagram



2.2 Actors & External Systems

Actor / system	Interaction	Protocol
Administrator, Manager, Employee, Custom role with `board.view`	View, filter and drag cards	HTTPS / JSON (Bearer JWT)
Task Management (TASK)	Board read, status change and assignee list endpoints; task scope; transition table; history	HTTPS / JSON
Project Management (PROJ)	Project list for the Project filter	HTTPS / JSON
Notification – reminder engine (NOTIF)	Stops reminding a task once a move makes it Completed	Shared database

No external (third-party) system is involved.

3. Architecture Goals, Constraints & Principles

3.1 Quality Goals

Goal	How the architecture supports it
Responsiveness	Optimistic move; filters applied in the SPA without reloading (NFR-BOARD-01, NFR-BOARD-03)
Correctness	Server is authoritative: permission, task access and transition table checked on every move; refused moves are rolled back on screen (NFR-BOARD-02, NFR-BOARD-04)
Consistency	One transition table with Task Details (NFR-BOARD-05)

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Goal	How the architecture supports it
Maintainability	No backend code of its own; reuses Task Management services
Compatibility	Drag-and-drop from the Angular CDK (NFR-BOARD-06)

3.2 Constraints

- Single API process and SQLite file (overall HLD).
- The board read is not paged: every visible task is loaded at once.
- Columns are fixed to the board statuses; cards are ordered by deadline and cannot be re-ordered.

3.3 Architecture Principles

- Reuse, do not duplicate: the board calls the Task Management endpoints and rules.
- The SPA never decides whether a move is allowed; it sends the move and follows the server's answer.
- Filtering only narrows data the user may already see.

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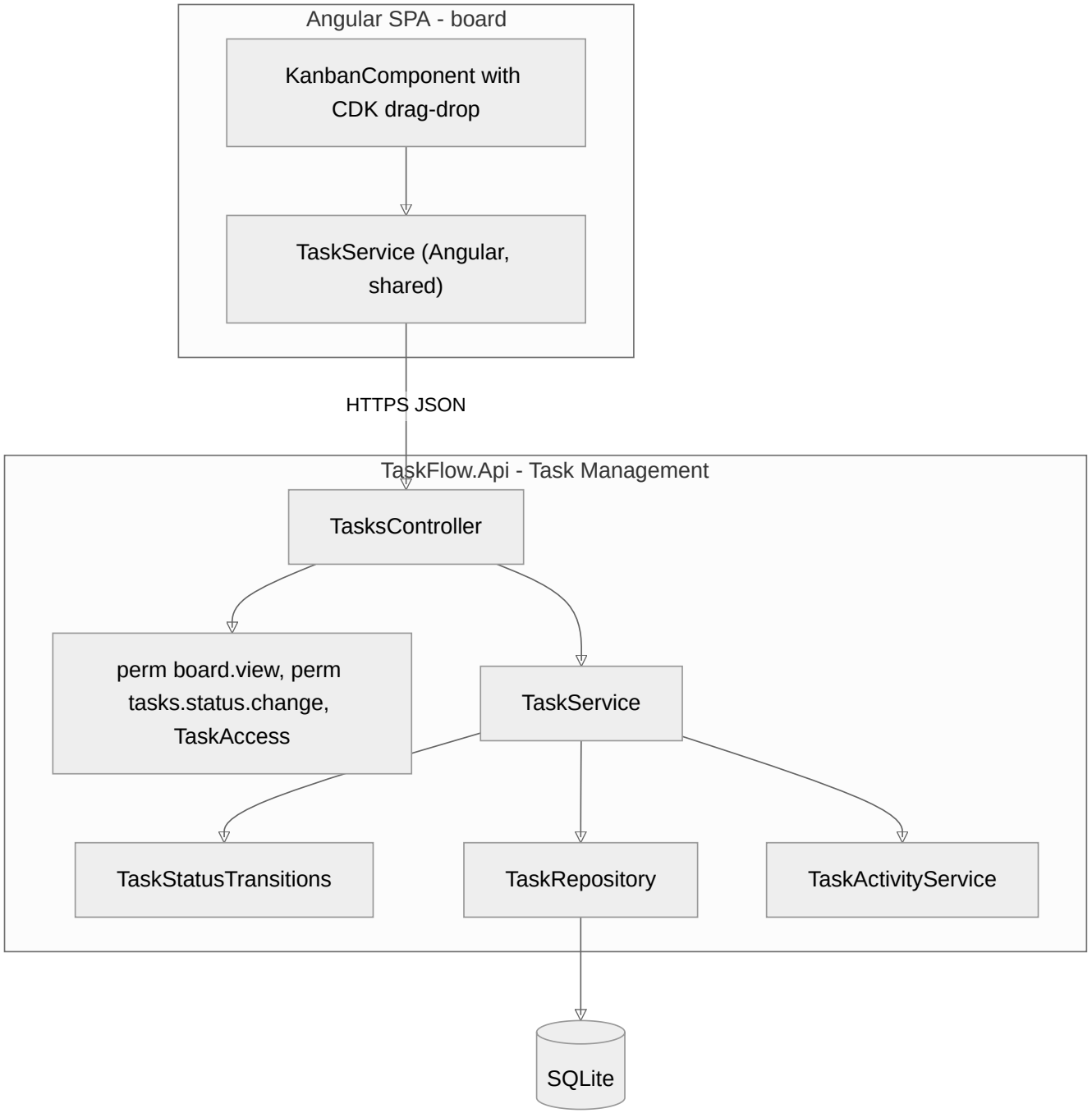
4. Architecture Overview

4.1 Architecture Style

The Board follows the system-wide style (Angular SPA + REST API, layered backend, SQLite via EF Core) described in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`). It is a frontend-heavy slice: one standalone page with client-side grouping, filtering and drag-and-drop, backed by three Task Management endpoints.

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4.2 Logical Architecture



4.3 Layers & Responsibilities

Layer	Responsibility (this sub-module)	May call
Presentation	Load, group into five columns, filter, count, highlight overdue, drag-and-drop with rollback, toasts, open card; `permissionGuard` on `/board`	Tasks API, Projects API
API / Controller	`TasksController` board, status and assignee actions (owned by Task Management)	`TaskService`

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Layer	Responsibility (this sub-module)	May call
Service	Task scope for the board read; transition check, time stamps and history on a move	`TaskRepository`, `TaskStatusTransitions`, `TaskActivityService`
Repository	All visible live tasks with assignee and project, ordered by deadline	Database (EF Core)
Integrations	None	–

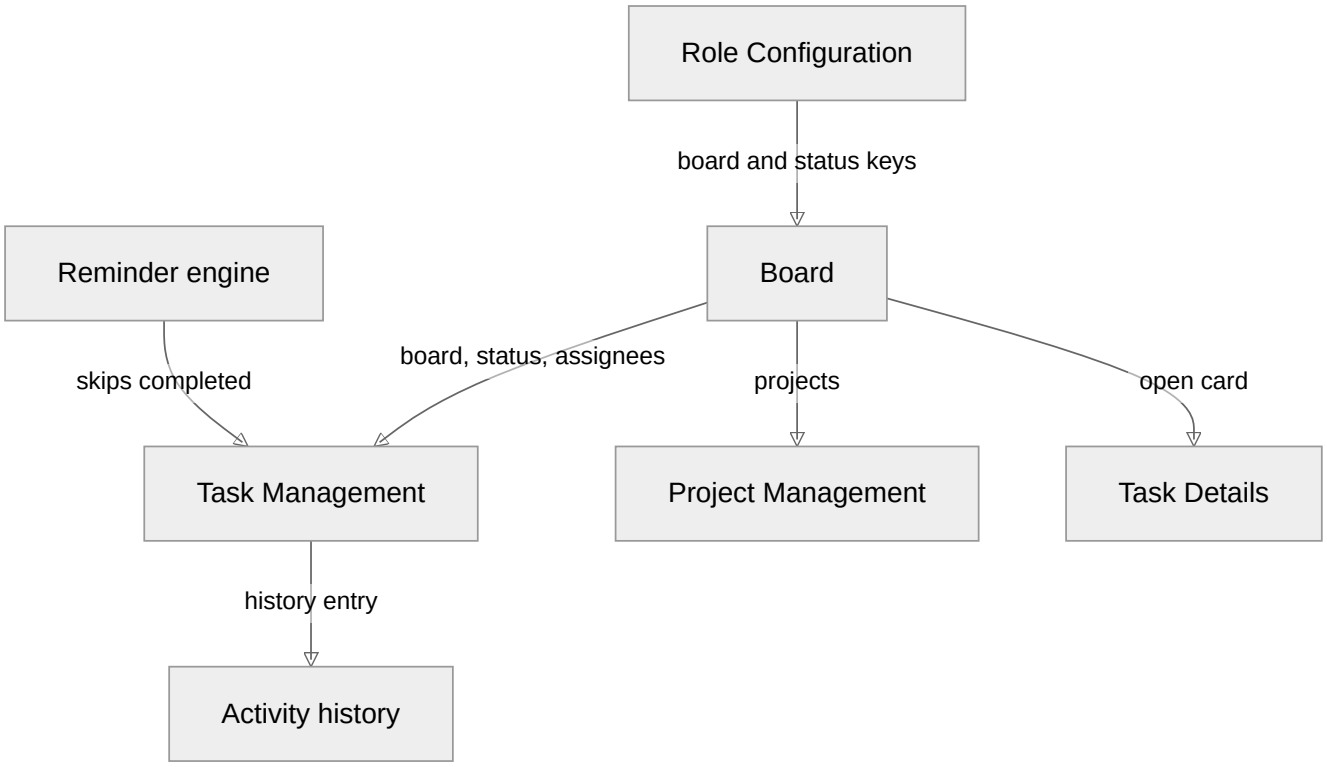
5. Component / Module Design

Component	Responsibility	Key components (high level)	Depends on
Board page (frontend)	Columns To Do, In Progress, In QA, Blocked, Completed; cards; filters; Refresh; drag-and-drop with optimistic update and rollback; navigation to Task Details with return to the board	`KanbanComponent`, `BoardColumn` model, `BOARD_STATUSES`	Angular `TaskService`, `ProjectService`, `AuthService` permissions, `@angular/cdk`
Board read (backend, shared)	Visible live tasks as cards with an overdue flag	`TasksController`, `TaskService`, `TaskRepository`	Task scope (Task Management)
Status change (backend, shared)	Permission, task access, transition table, completion / cancel time, history	`TasksController`, `TaskAccessHandler`, `TaskService`, `TaskStatusTransitions`, `TaskActivityService`	Task Management
Assignee list (backend, shared)	Distinct assignees of visible tasks for the Assignee filter	`TasksController`, `TaskService`	Task scope

API surface (high level): three actions under `/api/v1/tasks` (board read, status change, assignees), all owned and counted by Task Management; the board has no controller of its own. Detail in the LLD.

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5.1 Module Interaction



Partner	Direction	What is exchanged
Task Management	Out	Board read, status change and assignee list; task scope and transition rules
Project Management	Out (reads)	Project list for the Project filter
Task Details (Task Management)	Out (navigation)	Opens the task with a return URL to the board
Activity history (Task Management)	Indirect	"Status changed from X to Y" and "Task completed" written by the status change
Reminder engine (Notification)	Indirect	A task moved to Completed is no longer a reminder candidate
Role Configuration	In (claims)	`board.view`, `board.filters.use`, `tasks.status.change`, `tasks.view.all`

6. Data Architecture (high level)

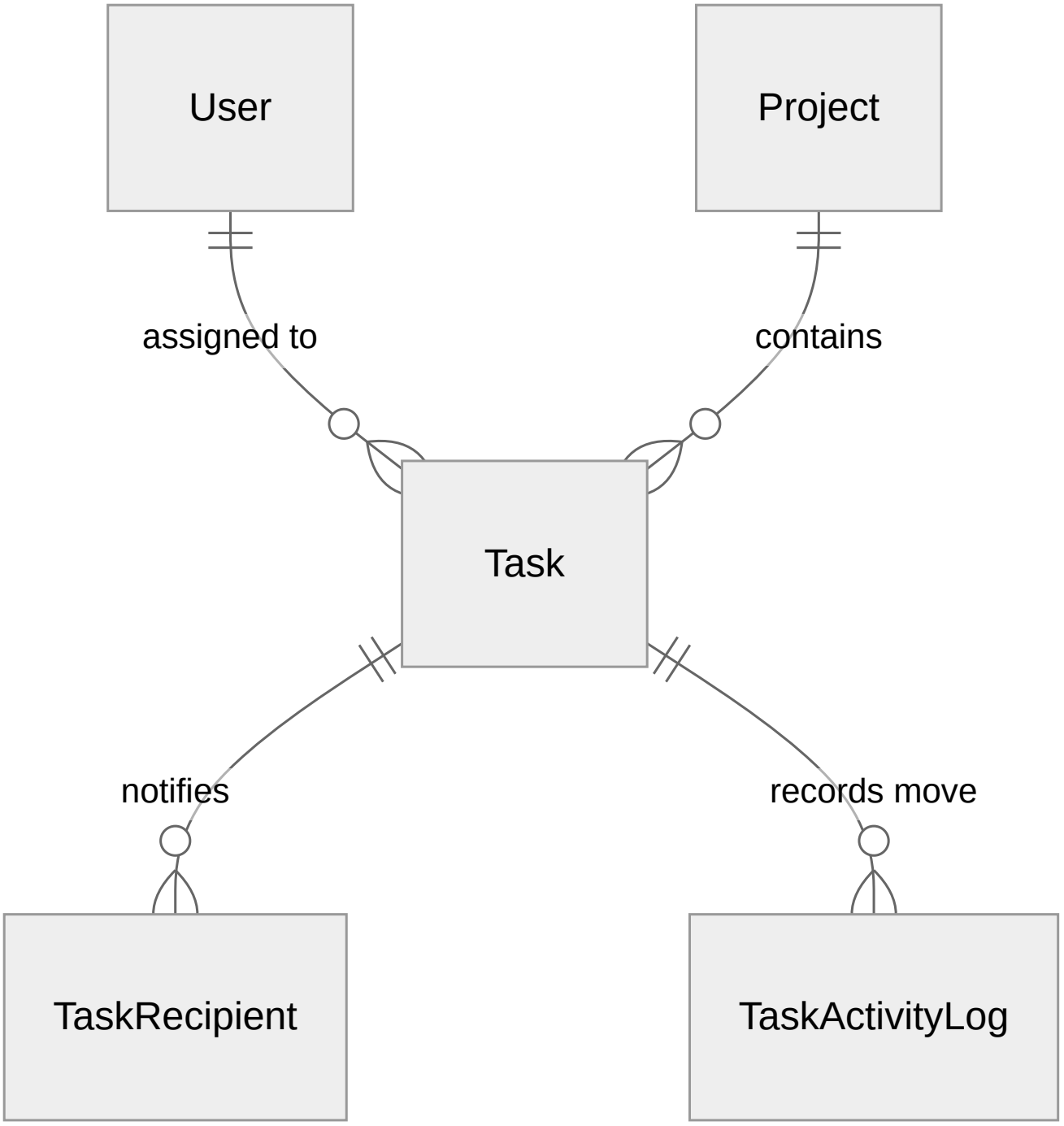
6.1 Data Stores

Store	Technology	Holds (this sub-module)	Notes
Main database	SQLite `taskflow.db`	Nothing owned; reads and updates `Task` status, appends `TaskActivityLog` through Task Management	Shared file; backup in the overall HLD
Browser memory	Angular signals	The loaded cards and filter state for the open page	Reloaded on open and Refresh

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Data domain	Owned / used
`Task`	Used and updated (status, completion / cancel time) via Task Management
`TaskActivityLog`	Appended via Task Management
`User`, `Project`	Used (card assignee and project names)
`TaskRecipient`	Used for the Own scope

6.2 Main Data Domains



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6.3 Data Flow

1. Open or Refresh: the SPA loads every visible live task as a card (with overdue flag) and, for view-all users, the assignee list; it loads the project list for the filter.
2. The SPA groups cards into the five board statuses (Cancelled cards are not shown), ordered by deadline, and applies the filters in memory.
3. A drop sends one status change; the server updates the task and appends history; the SPA keeps the new column or moves the card back.

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7. Integration Design

Integration	Direction	Purpose	Failure behaviour
Task Management board read	Outbound HTTPS	Cards in the caller's scope	"Could not load the board. Try again."
Task Management status change	Outbound HTTPS	Move a card	Card returns to its column; server reason or "Could not move {code}."
Task Management assignees	Outbound HTTPS	Assignee filter (view-all users only)	Filter stays empty
Projects API	Outbound HTTPS	Project filter options	Filter stays empty
Reminder engine	Indirect (database)	Stop reminders after Completed	Engine re-checks the task is still open before each delivery

No notifications or mail are sent for a move.

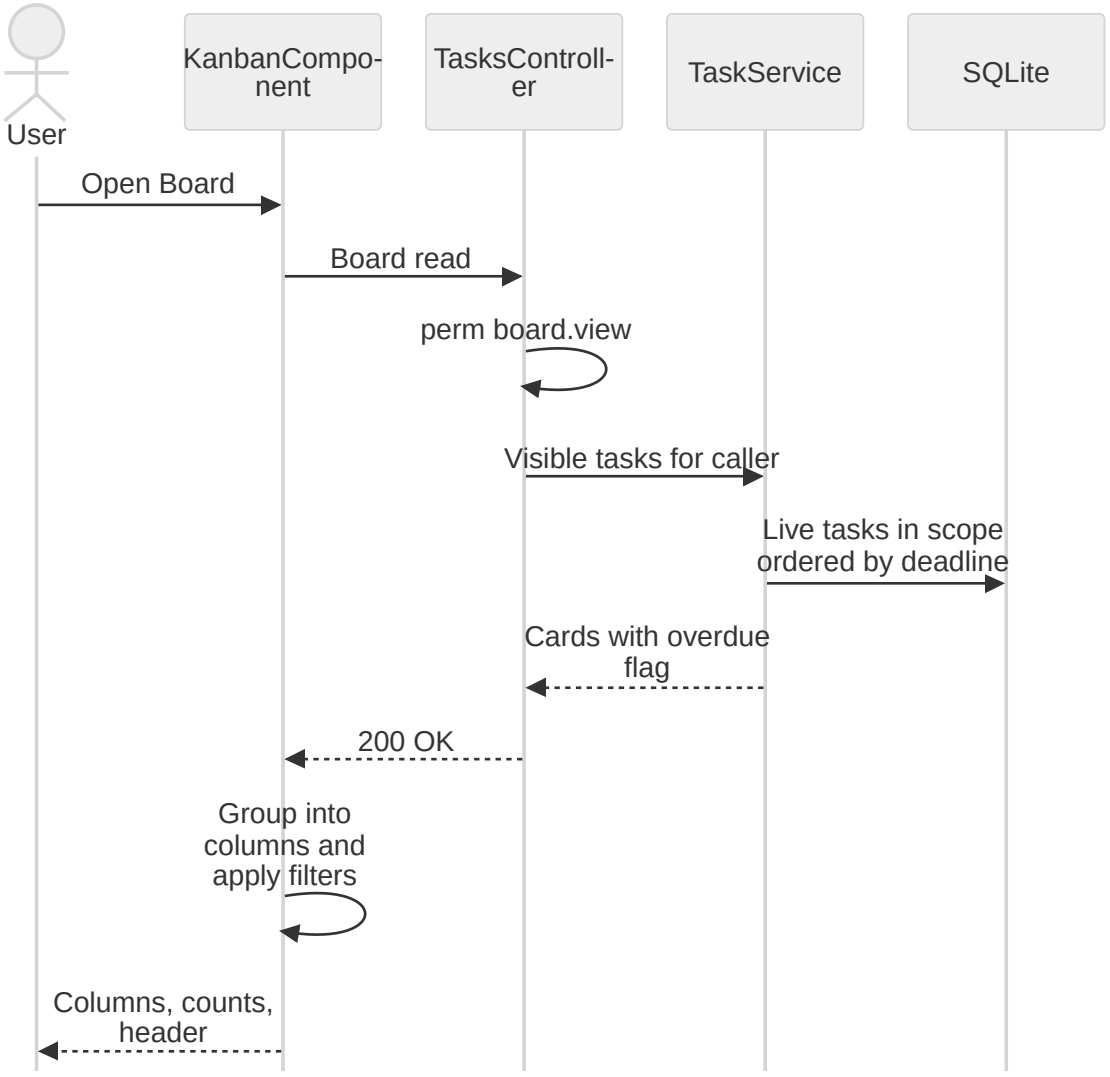
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8. Key Flows (high level)

Sign-in and the MFA gate are shared flows (overall HLD).

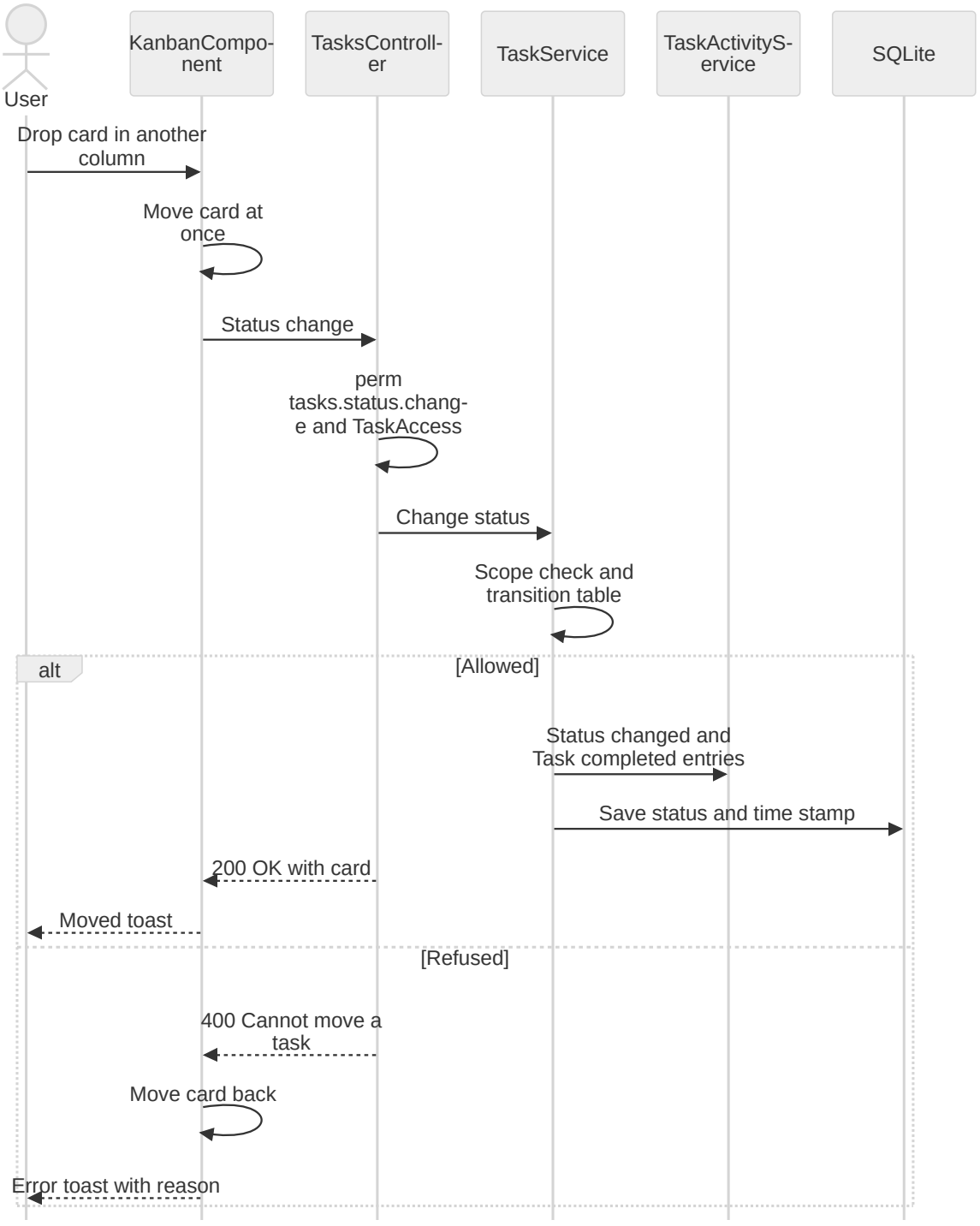
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8.1 Open the board



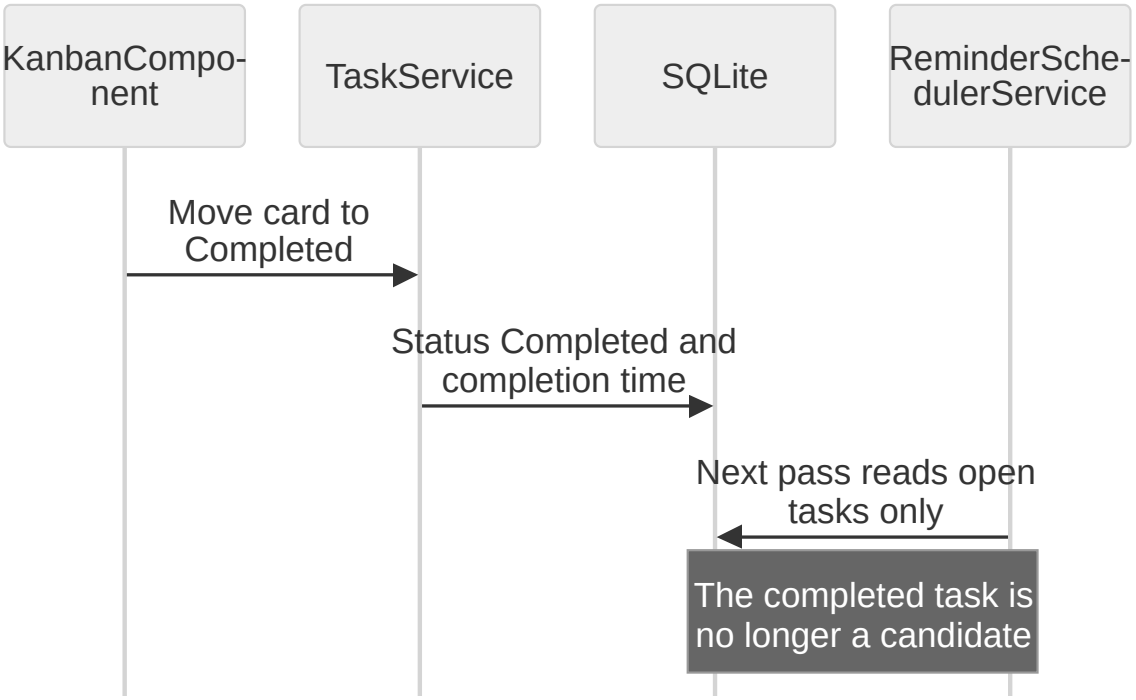
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8.2 Move a card (main business flow)



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8.3 Completed move and the reminder engine (background job)



9. Security Architecture

Concern	Approach
Authentication	Shared JWT bearer, session validation and mandatory MFA (overall HLD)
Authorization – board read	`board.view` policy; content limited to the caller's task scope (`tasks.view.own` always on, `tasks.view.all` for Team / Organization)
Authorization – move	`tasks.status.change` policy + `TaskAccess` resource policy + service scope check (Manager team) + transition table
Authorization – UI	`permissionGuard` on `/board` (`board.view`); dragging disabled without `tasks.status.change` ; filter row needs `board.filters.use` ; Assignee filter needs `tasks.view.all` ; menu item needs `board.view`
Data protection	Card data only (code, title, project, assignee, priority, deadline, status)
Auditing	Every accepted move is recorded in the task's activity history; refused authorizations logged by the denied-action audit
Known gaps	`board.filters.use` checked in the SPA only (filters are client-side); the Task List "Board view" button is not gated by `board.view` (§14)

10. Technology Stack

The sub-module uses the system stack (ASP.NET Core / .NET 10, EF Core 10.0.12 with SQLite, Angular ^19.2.0, RxJS ~7.8.0); versions and reasons are in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`). Specific to the board:

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Area	Technology	Version	Reason
Drag and drop	`@angular/cdk` (drag-drop)	^19.2.19	Accessible, maintained drag-and-drop lists for the columns
State	Angular signals (`@angular/core`)	^19.2.0	Cards, filters and computed column lists in memory

11. Deployment Architecture

The Board has no deployment unit of its own: it is a page of the Angular SPA and calls endpoints of the single `TaskFlow.Api` process. Hosting, environments and the data folder are described in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`).

12. Non-Functional Design

Area	Design decision
Performance	One load per open or Refresh; grouping and filtering in memory (NFR-BOARD-01); index on `Tasks(Deadline)` for ordering
Scalability	Not paged: all visible tasks are loaded (R-BOARD-01)
Reliability	Optimistic move rolled back on any refusal or failure, so the board never shows an unaccepted status (NFR-BOARD-04)
Availability	As the overall system
Maintainability	No backend code of its own; shared transition table (NFR-BOARD-05)
Usability	Immediate card movement, toasts, empty-column and filtered-empty texts (NFR-BOARD-03)
Observability	Request log per call; history entry per accepted move
Backup & Recovery	No own data; covered by the database backup

13. Design Decisions

ID	Decision	Alternatives considered	Rationale
D-BOARD-1	The board has no backend of its own; it uses the Task Management board read, status change and assignee endpoints	A separate board controller and service	One task scope and one set of status rules; less code
D-BOARD-2	Load all visible tasks unpaged and group / filter them in the SPA	Server-side paging and filtering per column	Instant filtering; simple API; acceptable for current volumes (R-BOARD-01)
D-BOARD-3	Optimistic move with rollback on refusal	Wait for the server before moving the card	Moves feel immediate while the server stays authoritative (NFR-BOARD-03, NFR-BOARD-04)

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ID	Decision	Alternatives considered	Rationale
D-BOARD-4	The SPA does not check transitions; the server's transition table decides	Duplicate the table in the SPA	One source of truth (NFR-BOARD-05); the refusal message comes from the server
D-BOARD-5	Five columns (To Do, In Progress, In QA, Blocked, Completed); Cancelled tasks are not shown	A Cancelled column	Cancelled is final and has no next status; the Task List Status filter finds them (R-BOARD-03)
D-BOARD-6	Overdue card flag computed on the server with the instant rule	Use the Task List day-based stage	Same overdue rule as the Dashboard and Reports; known difference with the list stage on the deadline day
D-BOARD-7	A move sends no notification; it is recorded in the activity history only	Notify recipients on each move	Avoids noise; history keeps the audit trail
D-BOARD-8	`board.filters.use` is enforced in the SPA only	Server-side filter check	Filters run on data already returned to the caller; they reveal nothing extra (R-BOARD-02)
D-BOARD-9	Cards are ordered by deadline and in-column re-ordering is ignored	User-defined card order	No order field to store; deadline order matches the Task List

14. Risks, Assumptions & Open Issues

Type	Description	Mitigation / owner
Risk	R-BOARD-01: large task volumes slow the board, because it is not paged	Use filters; add paging or server filtering later – Jojo George
Risk	R-BOARD-02: `board.filters.use` enforced only on the screen	Filters only narrow visible data – Jojo George
Risk	R-BOARD-03: Cancelled tasks are invisible on the board	Task List Status filter – Jojo George
Risk	R-BOARD-04: the Task List "Board view" button is shown without `board.view`	Gate the button – Jojo George
Assumption	Accessible task volume stays small enough to load at once	Revisit with paging – Jojo George
Open issue	The deadline filter compares dates in the browser time zone while overdue uses UTC now and the stage uses the application time zone	Align on the application time zone – Jojo George
Open issue	⚠️ To be confirmed: whether `TasksEndpointsTests.cs` covers the board read; no board-specific or frontend spec test is in the file list	Add board tests – Jojo George

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15. Traceability (requirements → components)

BRD/FRD item	Module / component	Section
FR-BOARD-01, FR-BOARD-02	Task Management board read (task scope) › `KanbanComponent` grouping into `BOARD_STATUSES`	§5, D-BOARD-1, D-BOARD-5
FR-BOARD-03, FR-BOARD-05, FR-BOARD-06	`KanbanComponent` cards, deadline order, header count	§5, D-BOARD-9
FR-BOARD-04	`TaskService` card overdue flag (instant rule)	D-BOARD-6
FR-BOARD-07	`KanbanComponent` CDK drag-drop, disabled without `tasks.status.change`	§9, §10
FR-BOARD-08	`TasksController` status change › `TaskAccessHandler` › `TaskService` › `TaskStatusTransitions`	§8.2, D-BOARD-4
FR-BOARD-09, FR-BOARD-10, FR-BOARD-11	`KanbanComponent` drop handling, optimistic move and rollback, toasts	§8.2, D-BOARD-3
FR-BOARD-12	`TaskService` status change › `TaskActivityService`	§8.2, D-BOARD-7
FR-BOARD-13, FR-BOARD-16, FR-BOARD-17	`KanbanComponent` client-side filters and empty states	§6.3, D-BOARD-2
FR-BOARD-14	`KanbanComponent` Assignee filter › Task Management assignee list (view-all only)	§7, §9
FR-BOARD-15	`KanbanComponent` filter row gating (`board.filters.use`)	§9, D-BOARD-8
FR-BOARD-18, FR-BOARD-19	`KanbanComponent` load and Refresh › board read	§8.1
FR-BOARD-20	`KanbanComponent` navigation to Task Details with return URL	§5.1
FR-BOARD-21, FR-BOARD-22	`board.view` policy, `permissionGuard` on `/board` , menu item placement	§9
NFR-BOARD-01 … NFR-BOARD-07	See §12	§12

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